

Multimodal Attribute-Integrated Person Re-Identification in UAV Surveillance Systems

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Abstract—Unmanned aerial vehicle (UAV) surveillance enables large-scale and flexible monitoring that surpasses conventional ground-based systems. Person re-identification (ReID) is a key UAV surveillance application for recognizing individuals across camera views and time. In addition, pedestrian attribute recognition (PAR) extracts high-level semantic attributes from individuals across camera views. However, UAV imagery presents challenges such as low resolution, varying viewpoints, and background interference, which limit the effectiveness of appearance-based ReID models. To address these limitations, this work proposes a multimodal ReID framework that integrates appearance-based features from ReID with semantic attribute features predicted by a PAR module. The ReID and the PAR models are based on Bag-of-Tricks (BoT) backbone and RegNet, respectively. Both feature types are combined through an early feature fusion strategy, followed by a re-ranking technique to enhance retrieval accuracy. Experiments conducted on the AG-VPReID dataset demonstrate that our attributed-integrated framework achieves 75.22% mAP and 78.83% Rank-1 accuracy, outperforming the performance of the baseline model. Furthermore, the t-SNE visualizations and silhouette scores validate that our framework yields feature representations with better separability compared to the baseline. Retrieval case analysis illustrates that the integration of attribute information enhances retrieval accuracy. Nonetheless, our framework continues to face limitations for poor image quality or individuals with highly similar attributes. Future research will focus on refining the framework to achieve higher accuracy and overall performance.

Index Terms—person re-identification, attribute integration, early feature fusion, multimodal data, UAV surveillance.

I. INTRODUCTION

Unmanned aerial vehicle (UAV) surveillance systems have emerged as a transformative technology, enabling large-scale and distributed monitoring that surpasses traditional ground-based systems. Unlike conventional surveillance with stationary viewpoints, UAVs provide diverse perspectives, superior maneuverability, and flexible deployment [1], making them a promising solution for extensive surveillance areas. Moreover, surveillance tasks are labor-intensive, repetitive, and costly

[2], which has driven the growing adoption of automated approaches using machine learning and computer vision.

Within UAV surveillance, crowd monitoring and tracking has become a primary focus, naturally leading to person re-identification (ReID)—the ability to recognize individuals across different camera views and time periods. In addition to ReID, pedestrian attribute recognition (PAR) extracts semantic attributes, such as demographic information or clothing style, from individuals across camera views. These attributes provide high-level and discriminative features that are robust to changes in viewpoint and environmental conditions [3]. Despite the potential, UAV imagery poses unique challenges, including low resolution, fluctuating altitudes, and varying viewing angles, making UAV surveillance significantly more difficult than in ground-based systems [4]. While most UAV research has concentrated on object detection and tracking, few studies have explored person ReID using UAV imagery [5], highlighting a critical gap in automated UAV surveillance.

Person ReID itself faces numerous challenges. Relying solely on appearance-based features often leads to suboptimal performance due to drastic viewpoint changes, misalignment, occlusions [6], varying lighting and illumination [7], and most importantly, background interference [7]. Models can become biased toward backgrounds rather than the person of interest, as backgrounds frequently occupy a substantial portion of the image. To address these issues, auxiliary information—such as attribute prediction, color extraction and image segmentation [6]—can be incorporated to enhance feature discrimination and improve ReID performance. [8].

Although ReID and PAR share the common goal of identifying individuals, they differ in focus. ReID emphasizes global but low-level appearance features, whereas PAR captures fine-grained and high-level semantic attributes. This distinction motivates our development of a multimodal ReID framework that integrates both appearance-based and semantic-based features as complementary sources of information. To achieve this, an early feature fusion strategy is

adopted, enabling the model to learn collaboratively from both feature types, rather than separately as in late feature fusion.

Our proposed framework first employs a ReID backbone based on Bag-of-Trick (BoT) [9], to extract appearance-based features from person images. Secondly, the PAR module built on RegNet [10] is constructed from the same images to predict semantic attributes such as gender, clothing style, and accessories. The dataset in this study is the AG-VPReID dataset [11]. Using an early feature fusion, the appearance- and attribute-based features are concatenated prior to training a final predictor model. To enhance retrieval accuracy, a re-ranking technique is applied. Finally, the performance of the attribute-integrated ReID model is compared with the baseline model. In addition, retrieval cases are analyzed to identify scenarios where the integrated model presents clear advantages as well as its limitations, thereby offering insights into how attribute-based features contribute to ReID performance.

Our paper is organized as follows. Section II reviews related work. Sections III and IV describe the proposed methodology and experimental setup. Section V presents the results and discussion, and Section VI concludes our work.

II. RELATED WORK

Early *person re-identification (ReID)* methods relied on hand-crafted unique features from person images, such as color histograms [12], texture descriptors [13], and geometric features [14], [15]. The introduction of convolutional neural networks (CNNs) significantly changed the field, achieving superior performance without requiring manual feature design [9], [16], [17]. This also led to benchmark datasets such as CUHK03 [17] and Market-1501 [18] datasets. Recently, research has extended from ground-based to aerial surveillance, with datasets such as PRAI-1581 [4], UAV-Human [19], and AG-VPReID [11] involving large-scale UAV imagery.

Attribute-assisted person ReID applies semantic attributes to enhance ReID, especially when appearance-based features are insufficient under challenging UAV imagery. Lin et al. [8] introduced the Attribute-Person Recognition (APR) network, jointly learning identity and attribute features, for pedestrian description at different granularities. AANet [20] integrated identity and attribute information, emphasizing discriminative attributes while suppressing irrelevant ones. Similarly, ReID-FSAI [21] combined semantic features with attribute recognition to improve matching accuracy in complex scenarios. Wang et al. proposed metric distillation [22], using attribute information to guide learning and providing interpretable explanations to humans. Extending this to aerial-ground ReID shows improvement in bridging different viewpoint domains. Nguyen et al. presented AG-ReID.v2 [23], addressing differences in viewpoints, poses, and resolutions between aerial and ground cameras and demonstrated that attribute information helped bridge the domain gaps.

Feature fusion techniques are commonly employed with multimodal data, with two strategies: late and early fusion [24]. In late fusion, each modality is processed independently, and the outputs are subsequently combined before the model

makes a final decision. While this late fusion approach offers flexibility and modality independence [25], it neglects cross-modal interactions [26]. In contrast, early fusion integrates all modalities at the input stage, enabling a unified model to be trained on the combined representation [26]. This early strategy facilitates the learning of joint representations and cross-modal relationships [27], making it effective when modalities provide complementary information. In the person ReID domain, APR [8] and AG-ReID.v2 [23] are examples early fusion approaches, whereas metric distillation [22] serves as a late fusion example.

III. METHODOLOGY

This section presents the dataset and the details of the proposed PAR-integrated ReID framework.

A. Dataset

The multimodal dataset in this work comprises person images and tabular pedestrian attribute data. The data is obtained from the AG-VPReID dataset [11], released for the AG-VPReID 2025 Competition. The dataset includes video-based images collected from three different sources: UAVs at various flying altitudes, CCTV and wearable cameras. Additionally, it provides tabular data of 15 soft-biometric attributes, such as gender, age, clothing style, and accessories.

B. Overall Framework

Our proposed PAR-integrated ReID framework consists of five main parts: the person re-identification (ReID) backbone, the pedestrian attribute recognition (PAR) module, the feature fusion, the backpropagation and the re-ranking. Figure 1 illustrates the first three main parts.

1) *Person Re-Identification (ReID) Backbone*: This backbone extracts features from person images and results in feature embedding vectors. Our ReID backbone builds on the Bag-of-Tricks (BoT) framework [9] that incorporates techniques such as warmup learning rate, random erasing, batch normalization and central loss to enhance the ResNet-50 backbone. Thus, the BoT framework outputs 512-dimensional image-based embedding vectors, denoted as $\{f_{img}\}$.

2) *Pedestrian Attribute Recognition (PAR) Module*: Both person images and attribute tabular data are used to construct the PAR module. Prior studies on UAV-based PAR [28] demonstrated that pre-trained CNN backbones are effective for extracting pedestrian attributes from UAV imagery. Our PAR module is based on RegNet [10], a TinyML model that has been shown to effectively extract features from UAV images and is well-suited for resource-constrained devices [29].

First, the person images are fed into RegNet, and its outputs are passed through a classification head with a sigmoid activation function, converting into probabilities of attributes. Next, these probabilities are inputs of a learnable embedding layer, which consists of three components:

- a. *Linear layer*: Converts the attribute probability vectors into a 512-dimensional feature representation, compatible with the outputs of the Re-ID backbone.

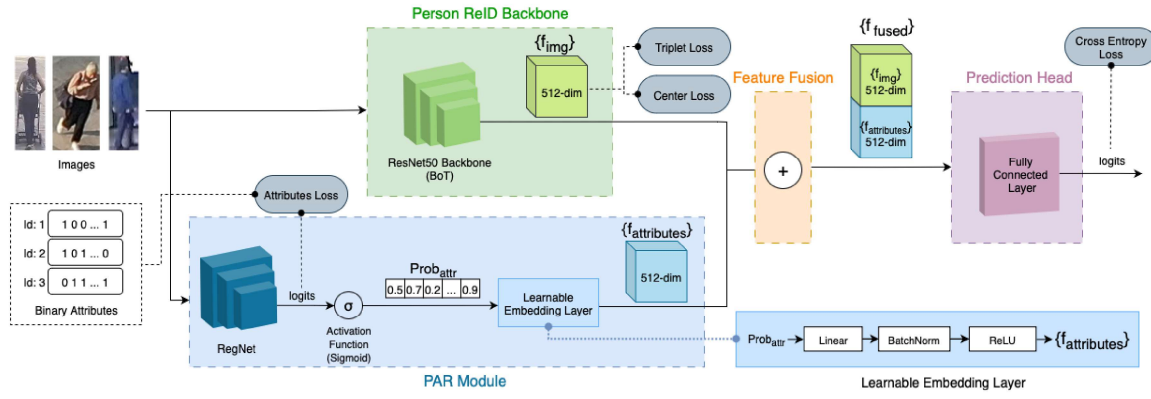


Fig. 1. Overview of our proposed PAR-integrated ReID framework.

- b. *BatchNorm layer*: Normalizes these representations to ensure that attributes with fewer classes (e.g., gender) do not have a disproportionately influence compared to attributes with more classes (e.g., clothing types).
- c. *ReLU activation*: Introduces non-linearity to the embedding features, allowing the model to capture more complex interactions between attributes (e.g., long hair + dress vs. short hair + sweater), instead of performing only a linear reweighting of attributes.

Consequently, the outputs of the learnable embedding layer in the PAR module results in the 512-dimensional attribute-based embedding vectors, denoted as $\{f_{attributes}\}$.

3) *Feature Fusion*: To obtain the PAR-integrated ReID model, an early feature fusion technique is applied, enabling the model to learn both feature types collaboratively rather than independently. The image-based vectors from the ResNet-50 backbone and the attribute-based embedding vectors from the PAR module are concatenated along the channel dimension to form a single 1024-dimensional fused feature, denoted as $\{f_{fused}\} = \{f_{img}, f_{attributes}\}$. This fused representation incorporates both global appearance information from the ReID backbone and the semantic attribute information from the PAR module. The fused vectors are fed into a prediction head, which consists of a fully connected layer that outputs the logits for each image.

4) *Backpropagation*: For each epoch, the total loss function of our proposed framework is computed as:

$$\mathcal{L}_{total} = \mathcal{L}_{Tri} + \alpha \mathcal{L}_{Center} + \beta \mathcal{L}_{Attr} + \mathcal{L}_{CE} \quad (1)$$

where α and β are weights of center and attribute losses respectively. The triplet loss ($\mathcal{L}_{Tri}$) for inter-identity separation and the center loss ($\mathcal{L}_{Center}$) for intra-class compactness are adopted from the BoT framework [9]. The attribute loss ($\mathcal{L}_{Attr}$) is formulated as cross-entropy loss computed from the attribute logits in the PAR module. Lastly, the cross-entropy loss with label smoothing ($\mathcal{L}_{CE}$) [9] is applied to the prediction head outputs. All losses from these components are back-propagated to optimize the model and enhance performance.

TABLE I
DETAILS OF TRAINING, QUERY, AND GALLERY SETS.

Data Set		#IDs	#Images
Train		383	793,056
Test	Query	165	330
	Gallery	165 + 7	8,200 + 350

5) *Re-Ranking*: Zhong et al. [31] introduced k-reciprocal encoding, assuming a gallery image in the k-reciprocal nearest neighbors of a query is more likely to be a correct match. This method encodes the k-reciprocal nearest neighbors of each gallery image into a single vector and re-ranks results. This refines the initial ranking and reduces false positives. Thus, it improves performance across various baselines and has become a standard approach in person ReID.

This work will evaluate a total of four models: (1) the BoT baseline framework, (2) the proposed PAR-integrated ReID framework, (3) the BoT baseline with re-ranking, and (4) the proposed PAR-integrated ReID framework with re-ranking.

IV. EXPERIMENTAL SETUP

This section describes the dataset preprocessing, implementation, and evaluation metrics used in our experiment setup.

A. Data Preprocessing

The AG-VPReID dataset is divided into training and test sets using a 70:30 ID ratio and ensuring that no ID appears in both sets. Since ReID is both a classification and retrieval task, the test set is further organized into query and gallery sets. Each set contains images of the same IDs but from different cameras. The gallery set includes seven distractor IDs (non-matching negative) to simulate real-world scenarios with non-target persons. The query set contains two randomly selected images per ID, while the gallery set comprises 350 images from seven distractors and 8,200 images from 165 IDs, with each ID represented by 14-50 images. Details of the training, query, and gallery sets are shown in Table I.

B. Implementation Details

For the ResNet-based ReID backbone, our implementation is based on the TorchReID repository [30]. The input images were resized to 256×128 pixels and augmented using random horizontal flipping and random erasing. The model was trained for 35 epochs using the AdamW optimizer (weight decay of 5×10^{-4} and an initial learning rate of 2×10^{-4}) with a two-stage scheduler: linear warm-up for five epochs, followed by cosine annealing (minimum learning rate of 1×10^{-6}). The batch size was 64. This configuration stabilized the early training phase and improved convergence.

During the derivation of the PAR-integrated ReID model, we froze the ReID backbone during the first five epochs to allow the PAR module to learn the meaningful attribute representations without being overshadowed by the stronger backbone gradients. This strategy encourages the PAR module to capture discriminative attribute features early, which can then be effectively fused with the ReID features once the backbone is unfrozen.

Experiments were conducted on an Intel i5-12400F CPU with 32 GB of RAM and an NVIDIA RTX 3060 (12 GB of VRAM), running Pop!_OS 22.04 LTS with Python 3.10.12. The models were implemented using PyTorch 1.2.1, TorchVision 0.16.1, and CUDA 12.1. The hyperparameters in the total loss function (1) were set as and $\alpha = 0.05$ and $\beta = 0.0005$.

C. Evaluation Metrics

Cumulative Matching Characteristics (CMC) is used as a basis for evaluation. CMC generally measures how well the model ranks the correct match in different positions.

Based on a query image set $\{p_1, p_2, \dots, p_N\}$, for a specific query image p_i and all L gallery images, the model outputs the corresponding fused feature vectors $f_{fused-p_i}$ and $\{f_{fused-g_1}, f_{fused-g_2}, \dots, f_{fused-g_L}\}$. Computing the pairwise cosine similarities between the query and the gallery vectors and sorting them in ascending order produces: $D_i = \{D_{(i,1)}, D_{(i,2)}, \dots, D_{(i,L)}\}$ and $D_{(i,1)} < D_{(i,2)} < \dots < D_{(i,L)}$, where $D_{(i,j)}$ is the cosine similarity between $f_{fused-p_i}$ and $f_{fused-g_j}$. If a correct match appears at the k^{th} rank in D_i , then this query achieves a top- k ranked result.

For performance evaluation, Rank-1 accuracy is used in this work, as it directly reflects the effectiveness of the proposed framework in retrieving the correct identity in the first attempt. Rank-1 accuracy is computed as the percentage of queries that receive a top-1 match.

In addition, the mean Average Precision (mAP) [18], capturing how well the correct matches are ranked overall, is used as another evaluation metric. The mAP is computed as:

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (2)$$

$$AP_i = \frac{1}{L} \sum_{j=1}^L r_j P(j) \quad (3)$$

where $P(j)$ is a fraction of correct matches in top- j results, and $r_j = 1$ for a correct match and 0 otherwise.

TABLE II
PERFORMANCE OF FOUR MODELS.

Model	Without Re-Ranking		With Re-Ranking		GFLOPs
	mAP (%)	Rank-1 (%)	mAP (%)	Rank-1 (%)	
BoT baseline	72.18	77.58	73.72	78.48	4.07
Proposed framework	73.12	80.06	75.22	78.83	4.34

V. RESULTS AND DISCUSSION

The performance comparison, the PAR integration effect, and the discussion insights are reported in this section.

A. Performance Comparison

Table II presents the performance of the four models: the BoT baseline and our proposed framework, each evaluated with and without re-ranking. Without re-ranking, the BoT baseline obtains a 72.18% mAP and a 77.58% Rank-1 accuracy, while our framework improves these to 73.12% and 80.06%, respectively. With re-ranking, the BoT baseline reached a 73.72% mAP and 78.48% Rank-1 accuracy, whereas our framework further achieves the performance to a 75.22% mAP and a 78.83% Rank-1 accuracy.

Table II compares the computational efficiency of the BoT baseline and our proposed framework. Our proposed model incurs a 6.6% increase in GFLOPs, achieving higher accuracy with only a modest rise in computation.

Although our proposed framework shows limited improvement over the baseline, the impact of integrating PAR into the BoT backbone will be analyzed further.

B. PAR Integration Effect

The t-Distributed Stochastic Neighbor Embedding (t-SNE) visualization is used to project the high-dimensional data (e.g. 512- or 1024-dimensional embedding vectors) into 2D space. When each vector represents a person image, a well-performing ReID model should produce t-SNE clusters where vectors of the same identity cluster together, while vectors of different identities are well-separated.

Figure 2(a) shows the embedding vectors from the BoT baseline model $\{f_{img}\}$, and Figure 2(b) shows the embedding vectors after integrating pedestrian attributes $\{f_{fused}\}$. Comparing the central regions in both figures, the embeddings with PAR integration exhibit better separability between identities and more compact clusters.

To quantify cluster compactness, the silhouette scores of $\{f_{img}\}$ and $\{f_{fused}\}$ are calculated and presented in Figure 3. A score closer to 1 indicates well-separated and compact clusters. The results demonstrate that embeddings from our PAR integrated ReID framework provide better identity separation than those from the BoT baseline, even at early stages. Although the silhouette scores of both models become more similar as training progresses, the PAR-integrated ReID model consistently outperforms the BoT baseline, demonstrating that attribute integration enhances identity discrimination.

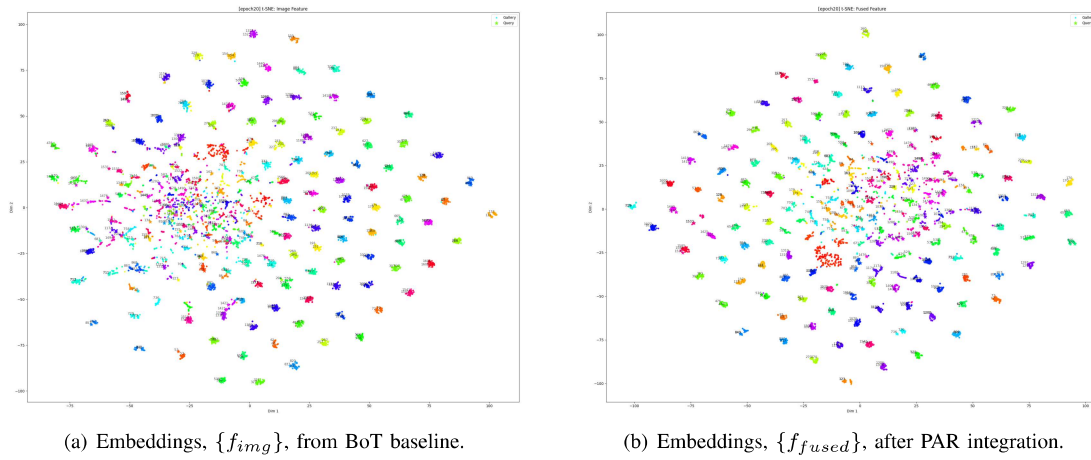


Fig. 2. t-SNE visualizations of embedding vectors.

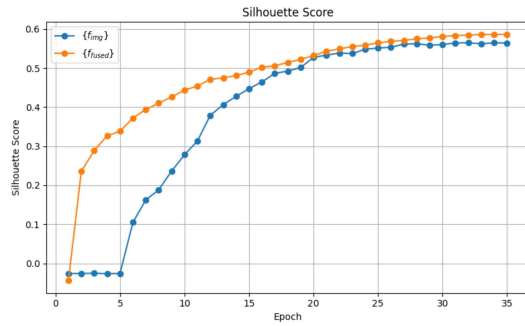


Fig. 3. Cluster performance: silhouette scores.

C. Discussion

Figure 4 illustrates examples of results where PAR integration guides the model toward more correct retrievals. The BoT baseline, relying on appearance similarity for both person and background, can produce incorrect matches. For instance, the BoT baseline results in Figure 4 exhibit incorrect matches partly due to background similarities between the query and the gallery images. By incorporating pedestrian attributes, the ReID performance is enhanced. For example, the results from our proposed framework in Figure 4(a) show a case with no match for a hat, whereas Figure 4(b) shows more correct matches for a bald hairstyle. These examples show that the semantic information from attributes—such as gender, hairstyle, and clothing—complements appearance-based features, providing additional context and improving accuracy.

Additionally, we analyze the primary causes contributing to incorrect retrievals in our proposed framework. Two main causes are identified. The first is poor image quality, as shown in Figure 5(a). Low-resolution or low-light conditions limit the model’s ability to accurately recognize and classify attributes. The second cause arises when different individuals share very similar attributes as shown in Figure 5(b). This similarity makes it more challenging for the model to distinguish between them. Moreover, it suggests that the model

tends to rely more on global appearance-based features rather than fine-grained and distinctive characteristics. Our future work will focus on reducing these errors, particularly the latter cause, to further enhance retrieval accuracy.

VI. CONCLUSION

In this paper, a multimodal framework for person re-identification in UAV surveillance that integrates appearance-based features with semantic attributes is proposed. The framework combines a BoT backbone with a RegNet-based PAR module using an early fusion. Experimental results on the AG-VPReID dataset show that our proposed framework achieves 73.12% mAP and 80.06% Rank-1 while adding only a 6.6% computational overhead compared to baseline. The t-SNE analysis and silhouette scores also confirm that our PAR-integrated framework yields better identity separation and more compact feature clusters. Although limitations remain in cases of low-quality images and individuals with similar attributes, this work demonstrates the potential of multimodal attribute-integrated approaches for enhancing person ReID, contributing to more effective automated UAV surveillance systems. Future work will focus on optimizing the framework with advanced fusion techniques and incorporating error analysis to enhance its real-world relevance.

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Fig. 4. Comparisons between retrieval results from the BoT baseline and our proposed framework. Images bounded by green boxes indicate correct matches, while images bounded by red boxes denote incorrect matches.



Fig. 5. Examples of incorrect retrievals.

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